

Deploying and Monitoring an ML Workflow

Learning Objectives

- Understand the challenges Nova HealthTech faced in managing code updates, including inconsistencies, delays, and scaling issues.
- Learn the concept and components of CI/CD, including Continuous Integration, Continuous Deployment, and automated testing.
- Explore how version control systems, integration tests, regression tests, and staging environments are integrated into a CI/CD pipeline.
- Study the process of building a CI/CD pipeline tailored to MLOps needs, emphasizing automation and reliability.
- Examine the benefits of CI/CD in improving update speed, system reliability, and collaboration efficiency

Agenda

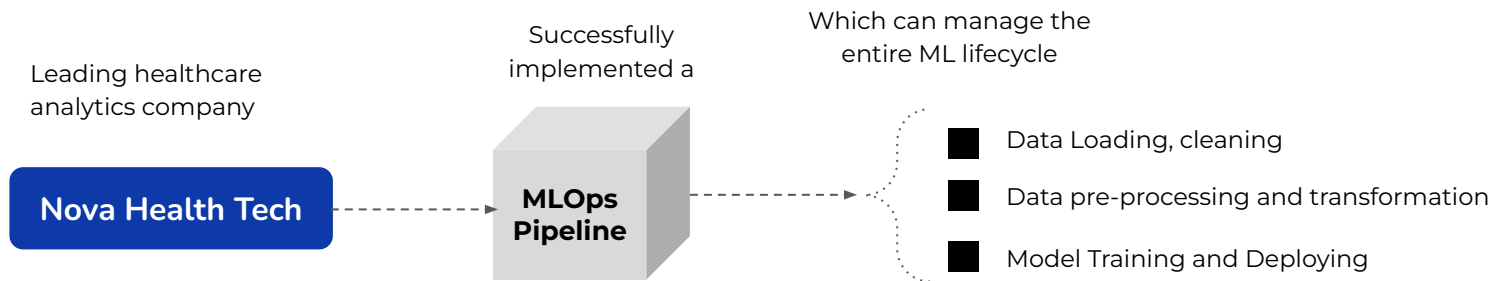
- Business Context and Problem Statement
- Challenges and Requirements
- Introduction to CICD
- Why CICD?
- How CICD works?
- Benefits of CICD
- CICD on Azure with Github Actions
- Conclusions

Business Context and Problem Statement

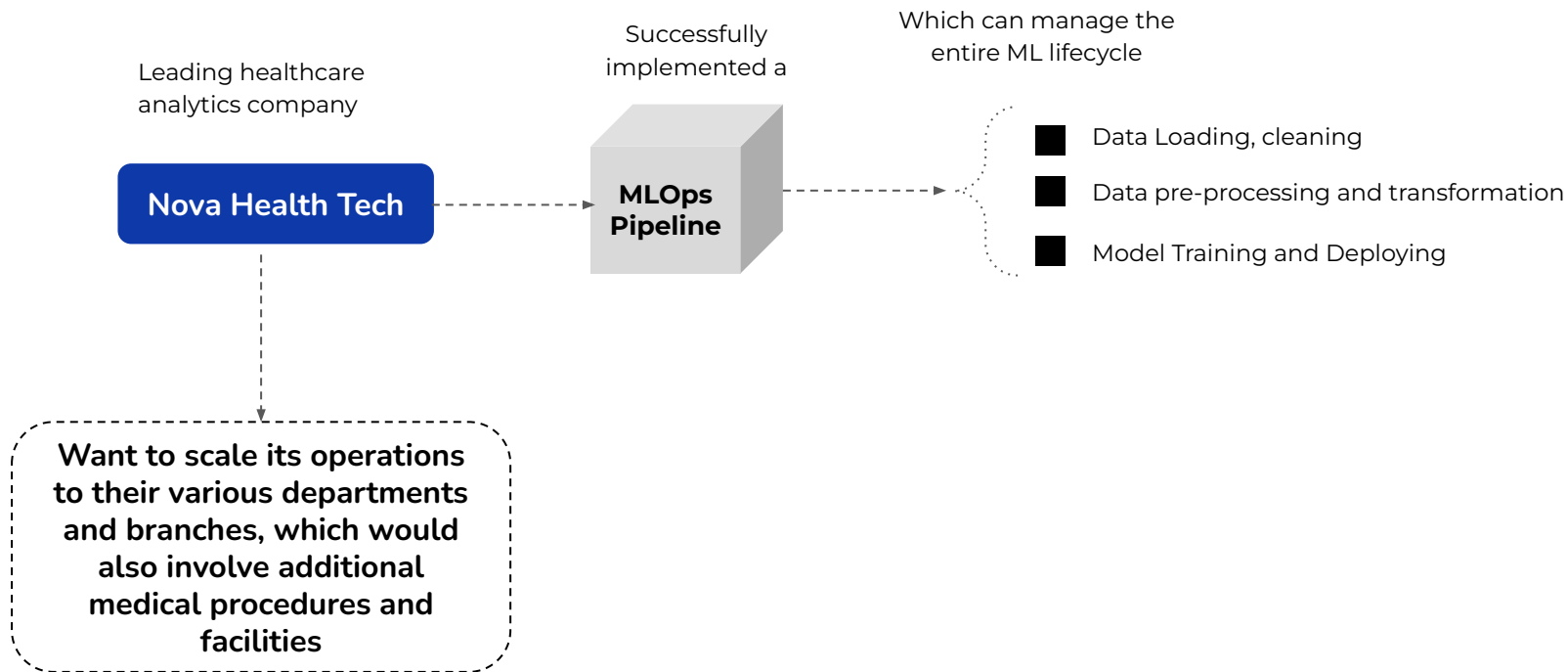
Leading healthcare
analytics company

Nova Health Tech

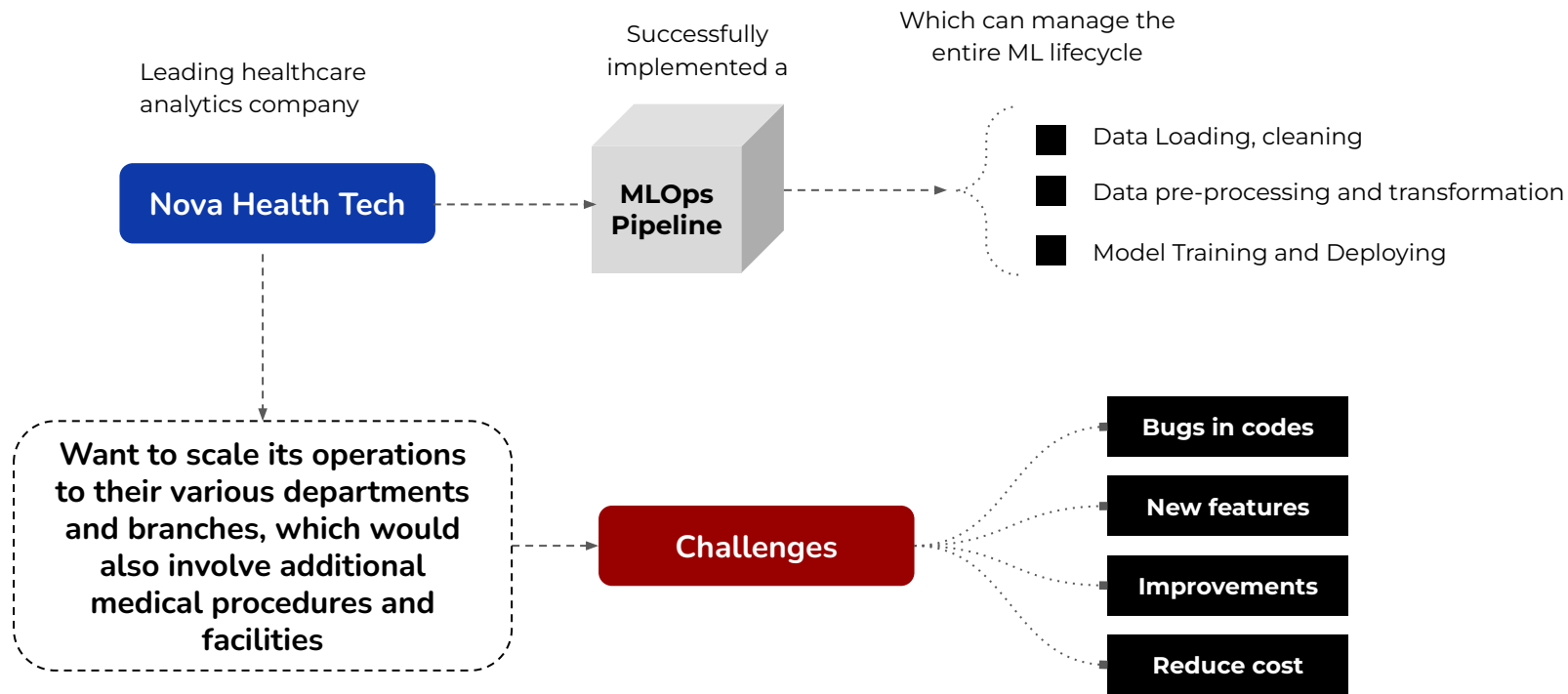
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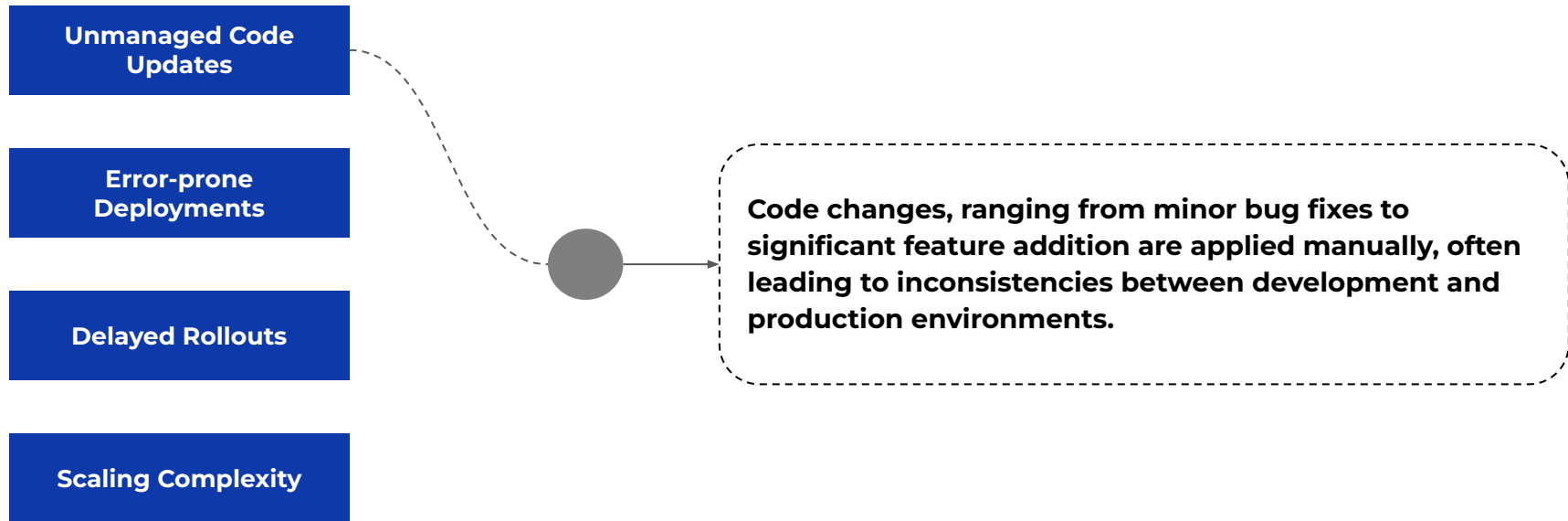
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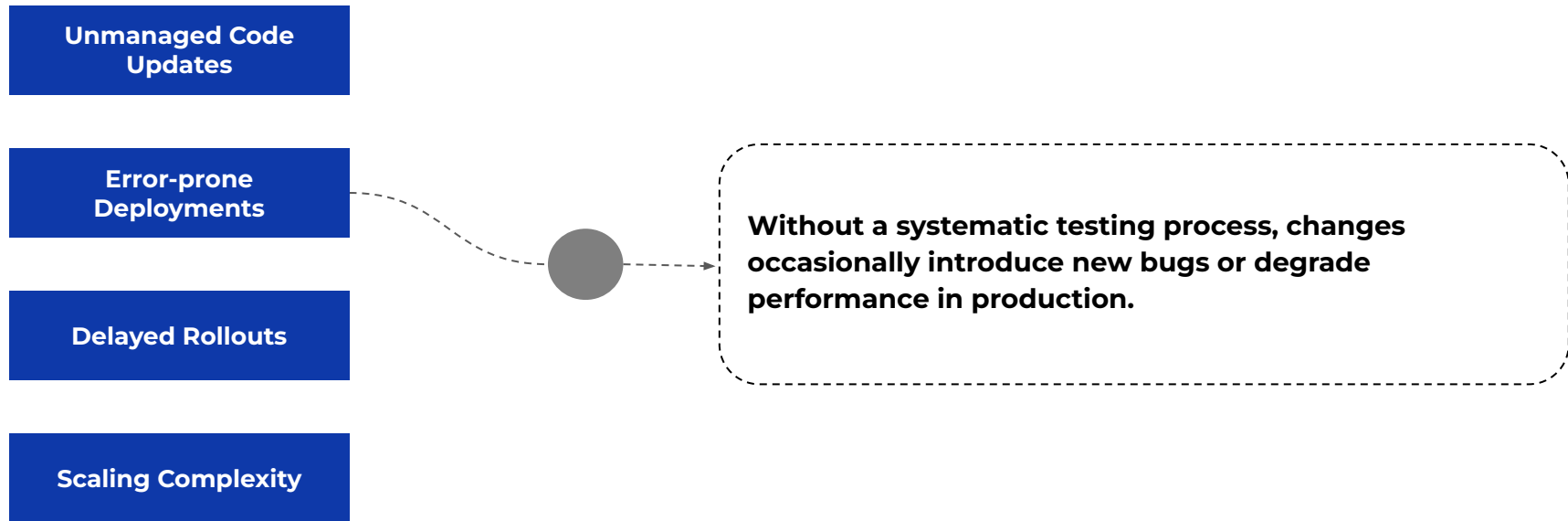
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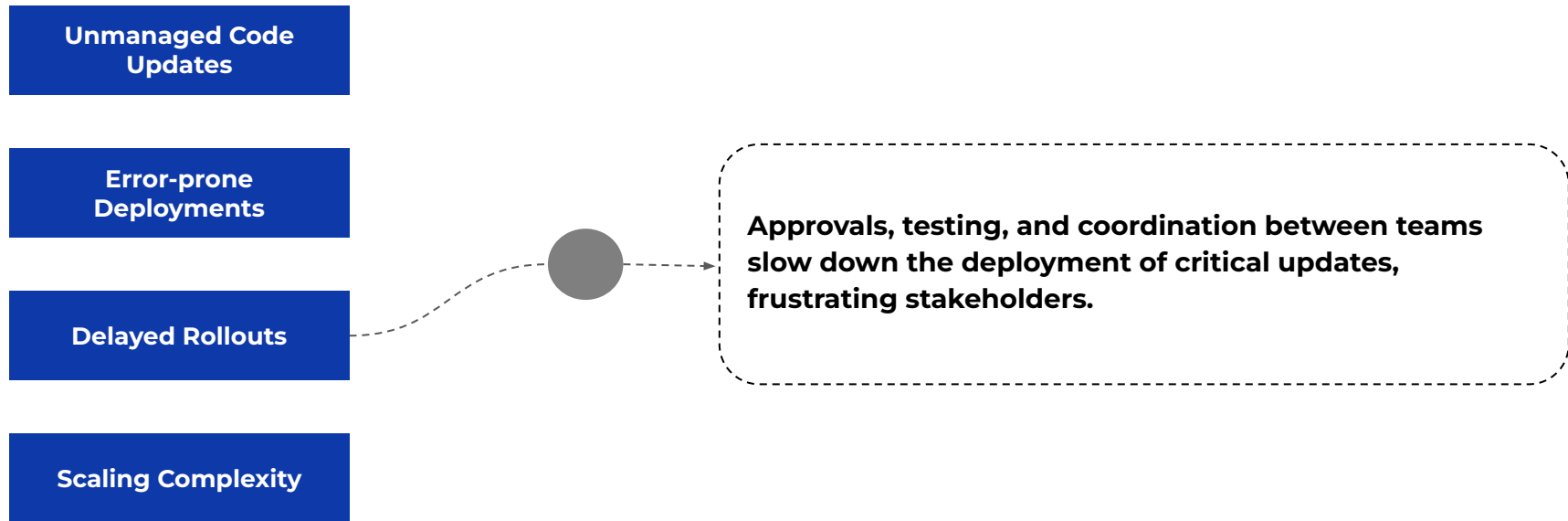
Challenges



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Unmanaged Code Updates

Error-prone Deployments

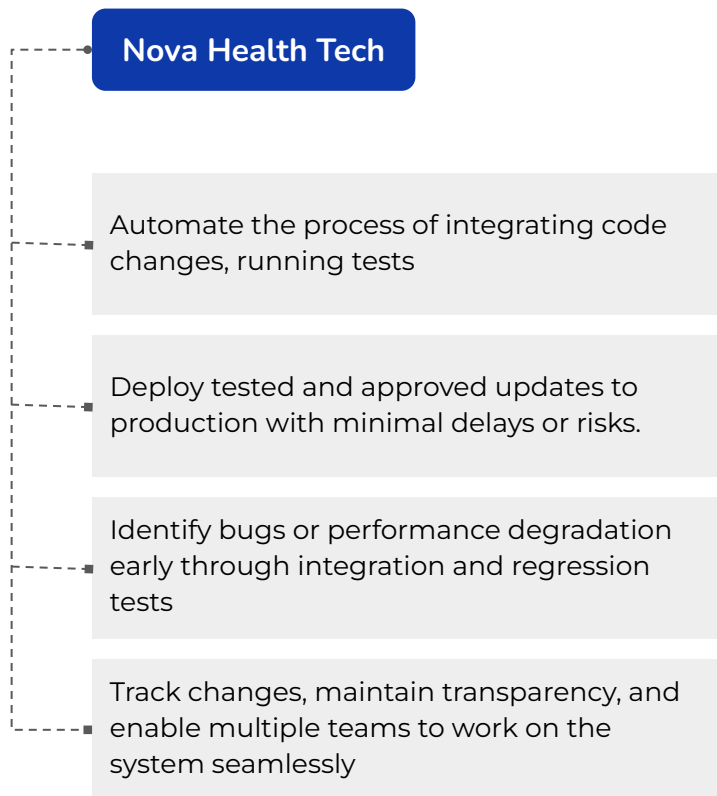
Delayed Rollouts

Scaling Complexity

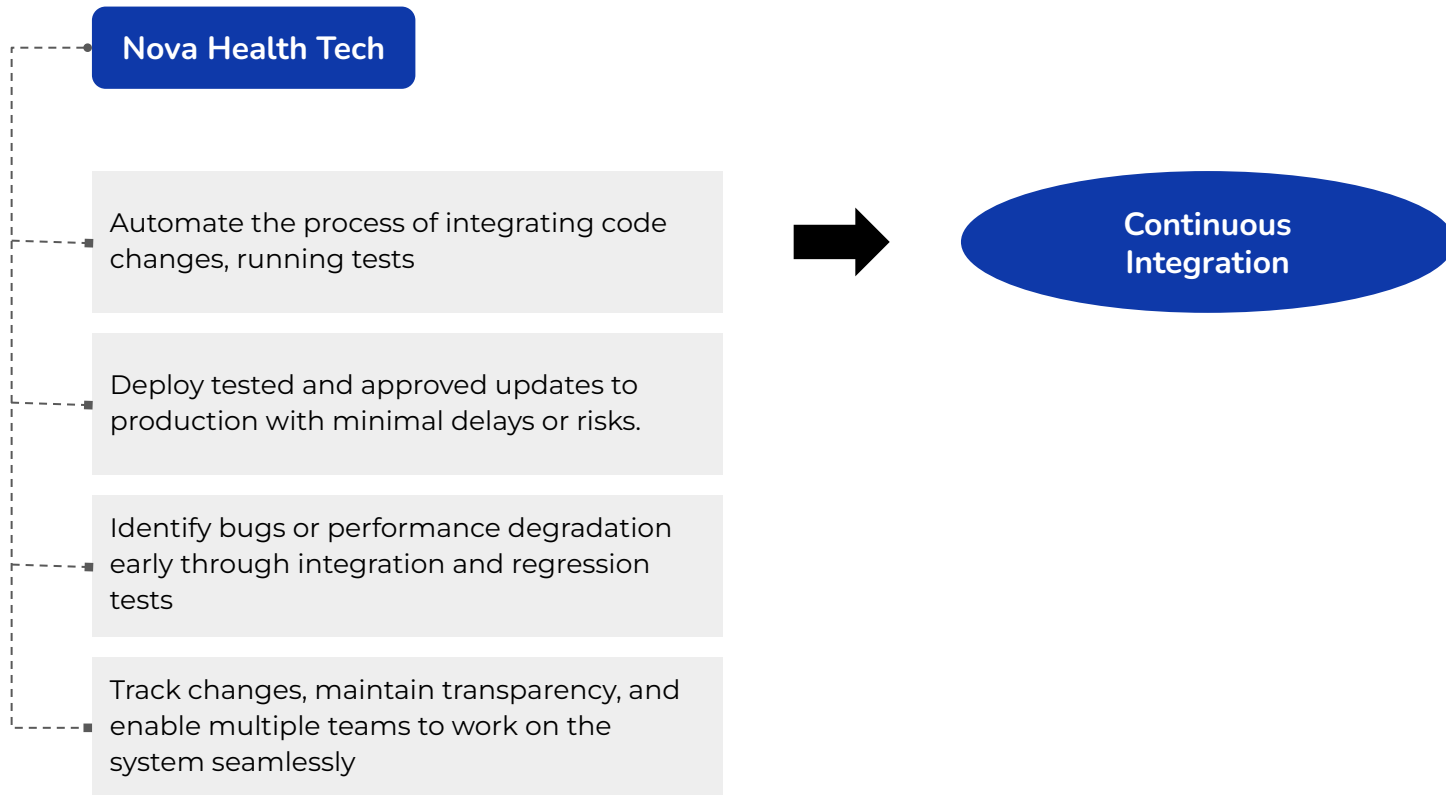


As more hospitals adopt the system, Nova needs a scalable process to quickly and reliably apply updates across multiple environments

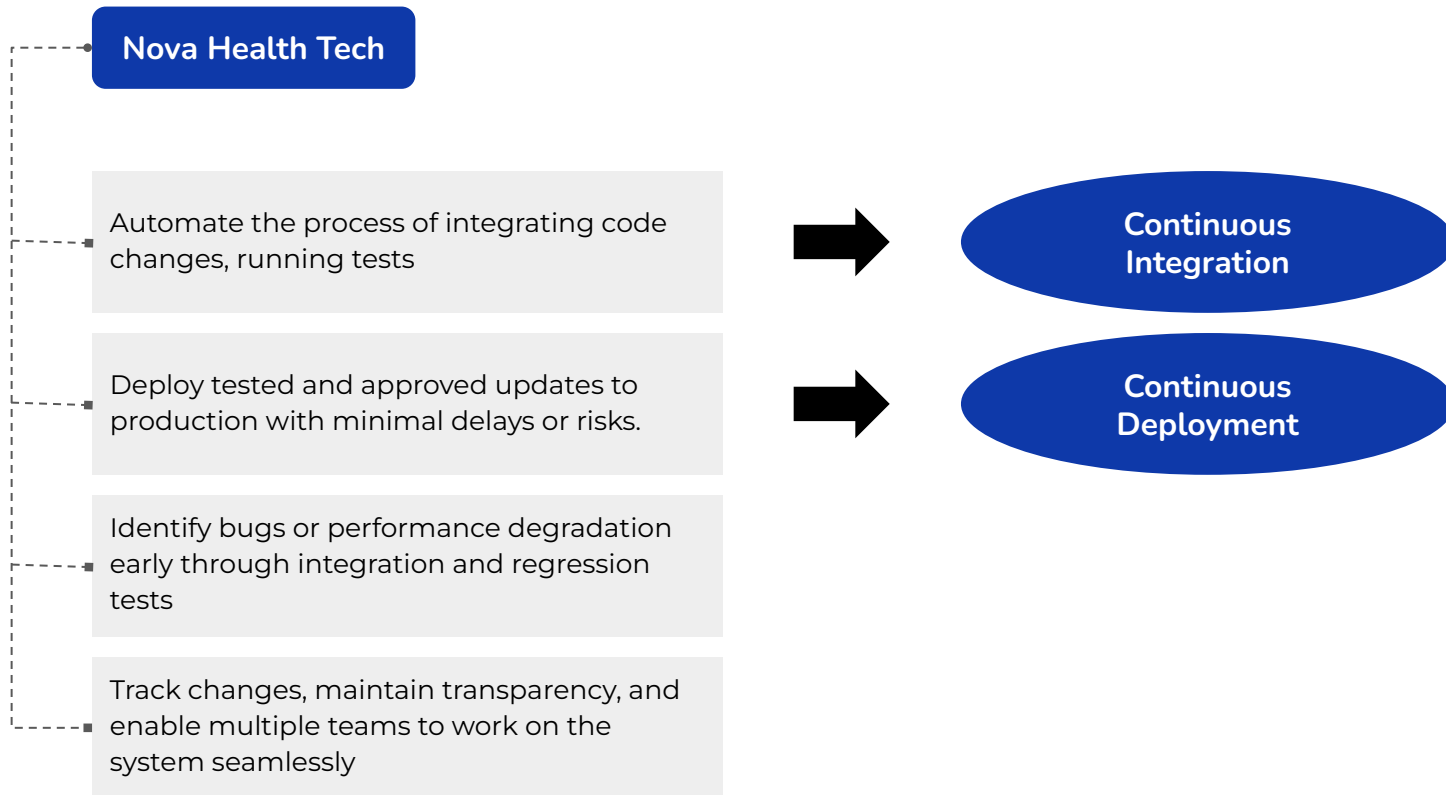
Needs of Nova Health Tech



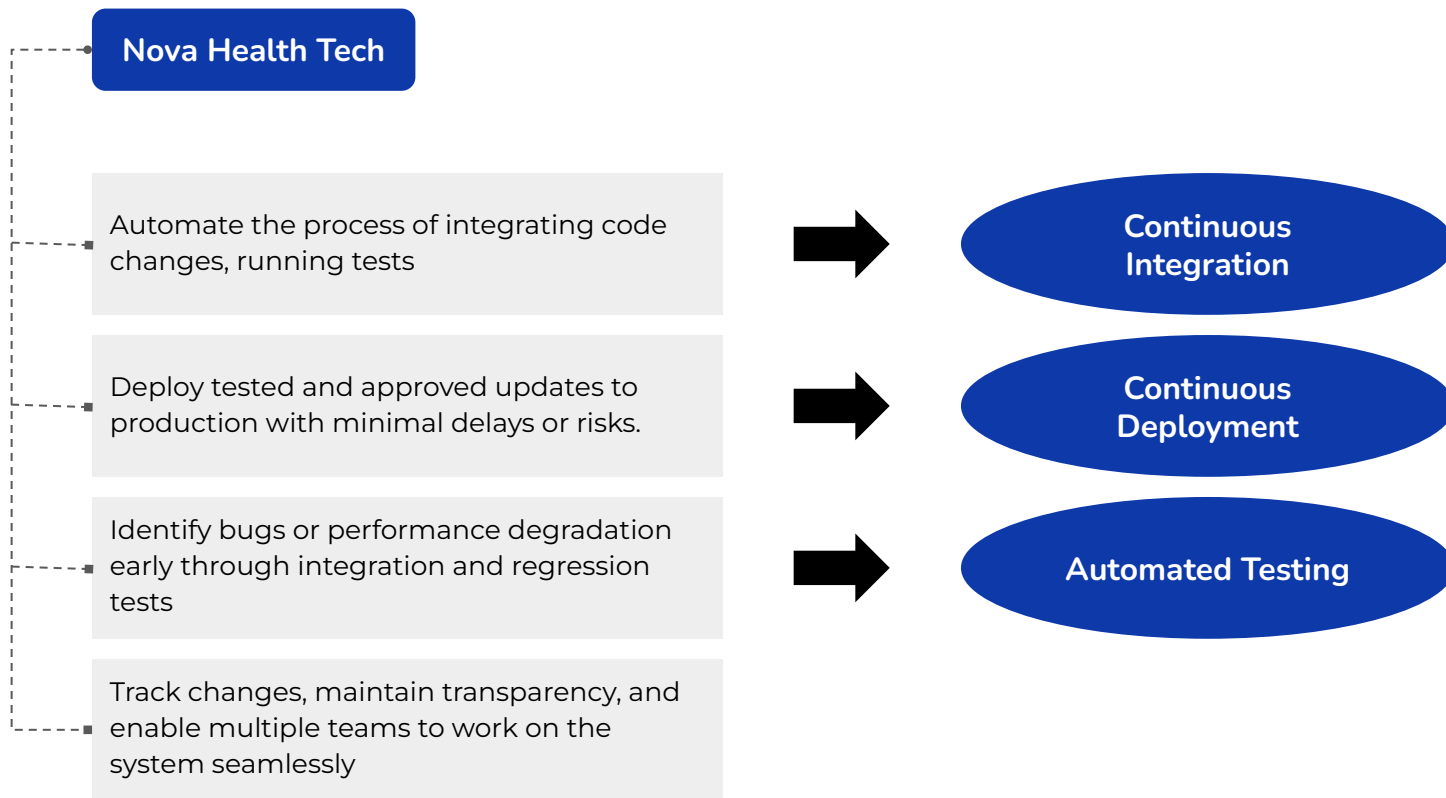
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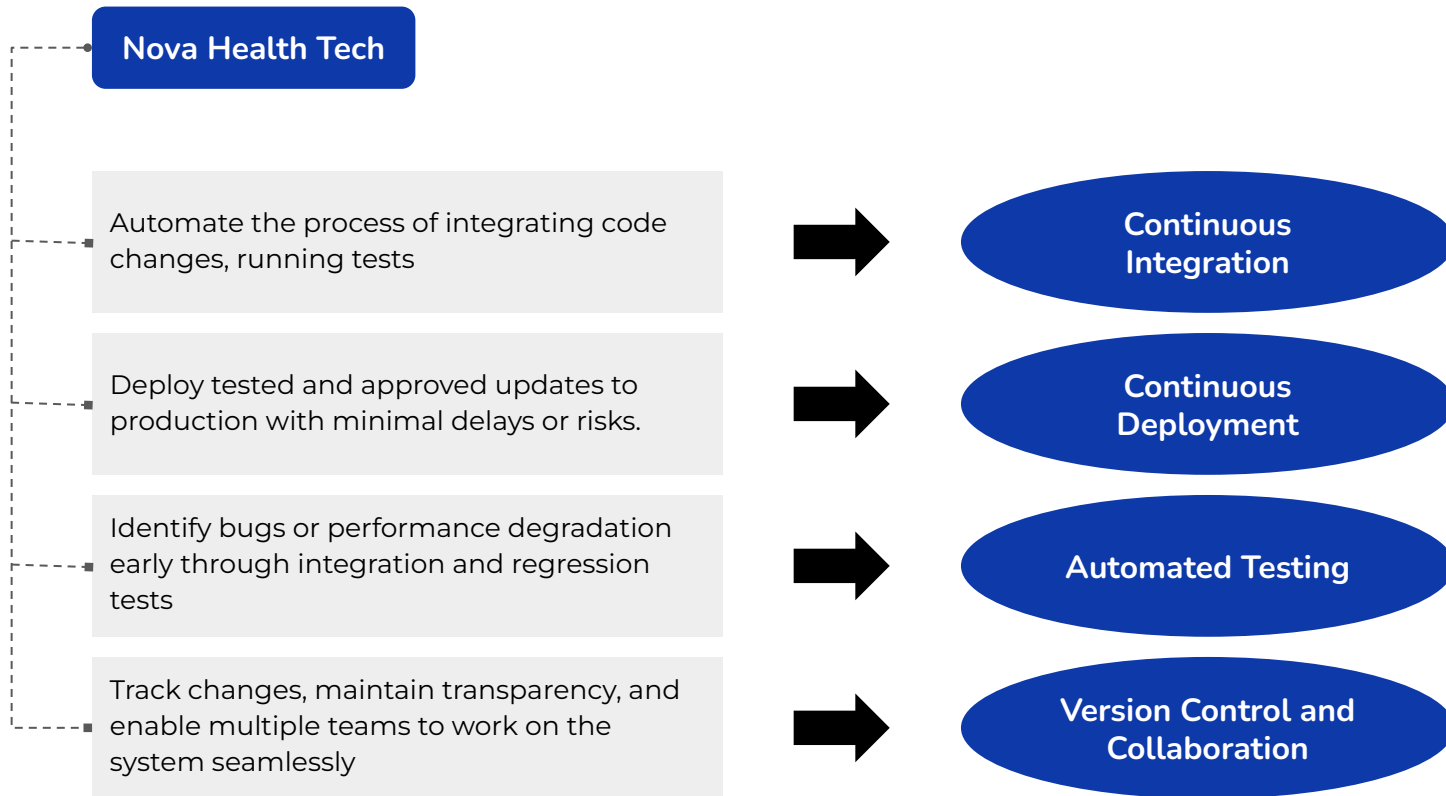
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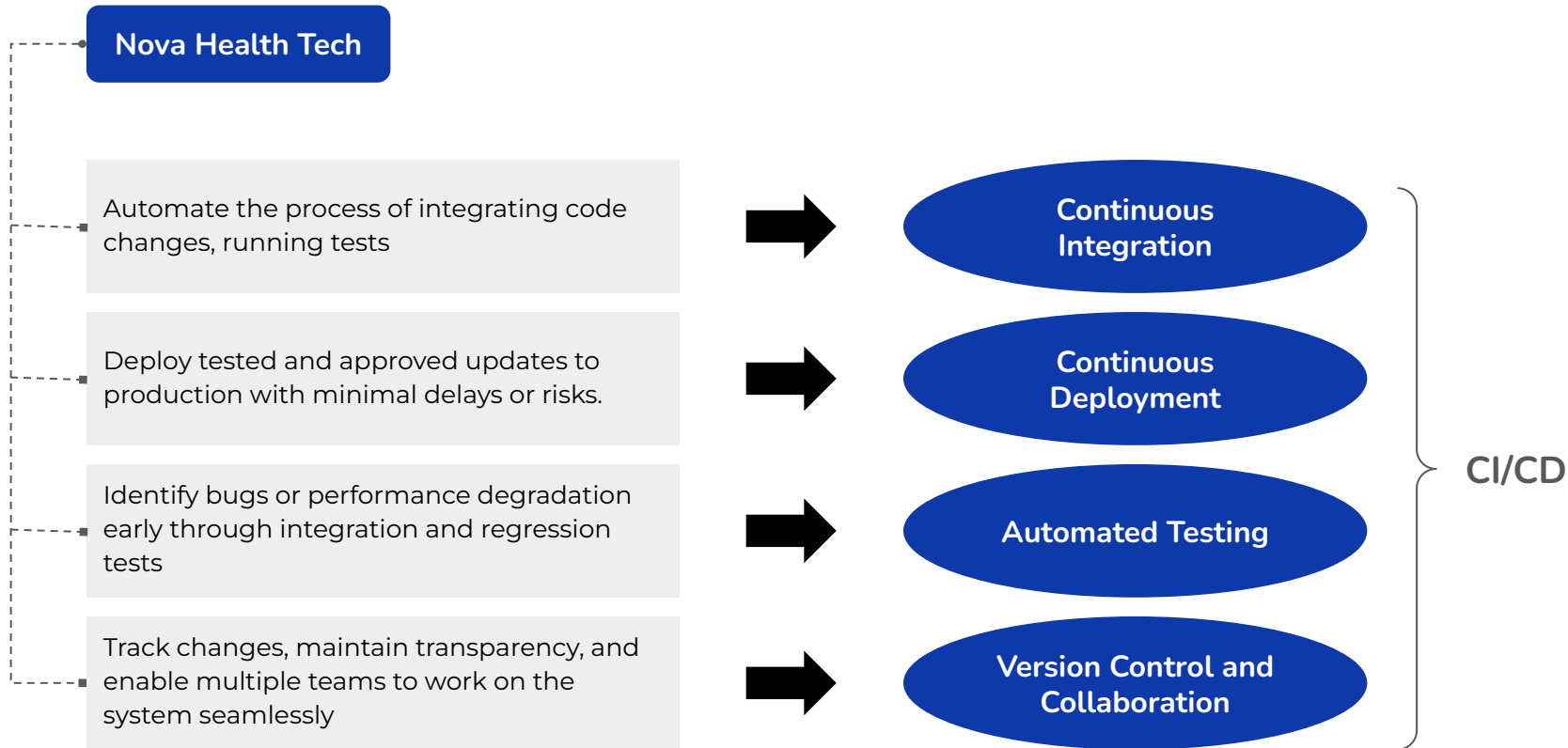
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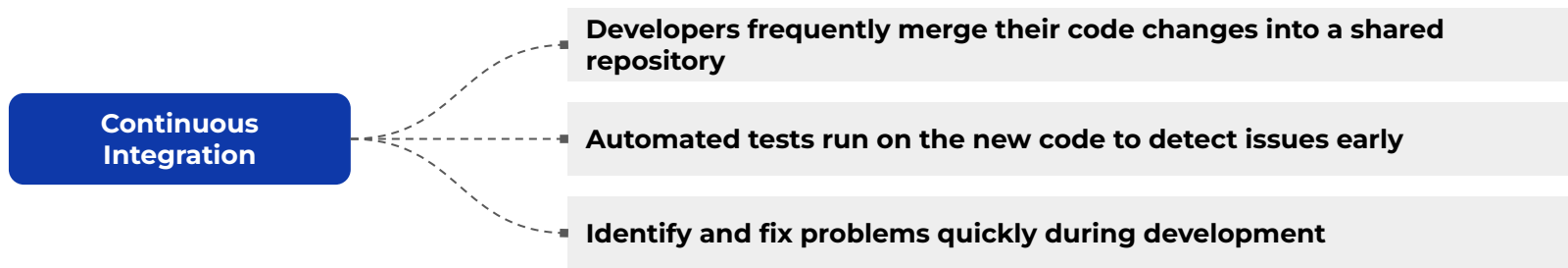


Introduction to CICD

CICD stands **C**ontinuous **I**ntegration and **C**ontinuous **D**eployment - automates and streamlines the software development lifecycle

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Continuous Integration

- Developers frequently merge their code changes into a shared repository
- Automated tests run on the new code to detect issues early
- Identify and fix problems quickly during development

Continuous Deployment

- Automates the process of delivering code to production after passing all tests
- Ensures that every validated change is deployed to users
- Deliver features and fixes faster with minimal manual intervention

How CI/CD Works?

1

Version Control

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- Additional test suites validate the code, including unit, integration tests

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Deployments

- Once tests pass, the code is automatically deployed to staging or production environments.
- Monitoring tools ensure the deployment works as expected.

Benefits of CICD

Faster Rollouts

Updates, whether bug fixes or new features are deployed within hours instead of weeks.

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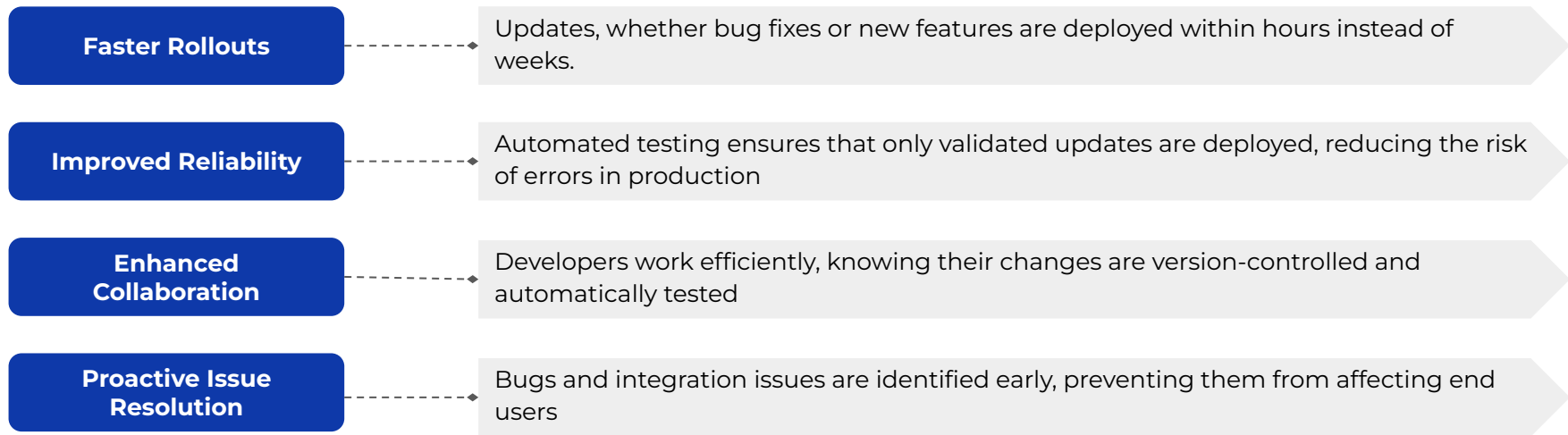
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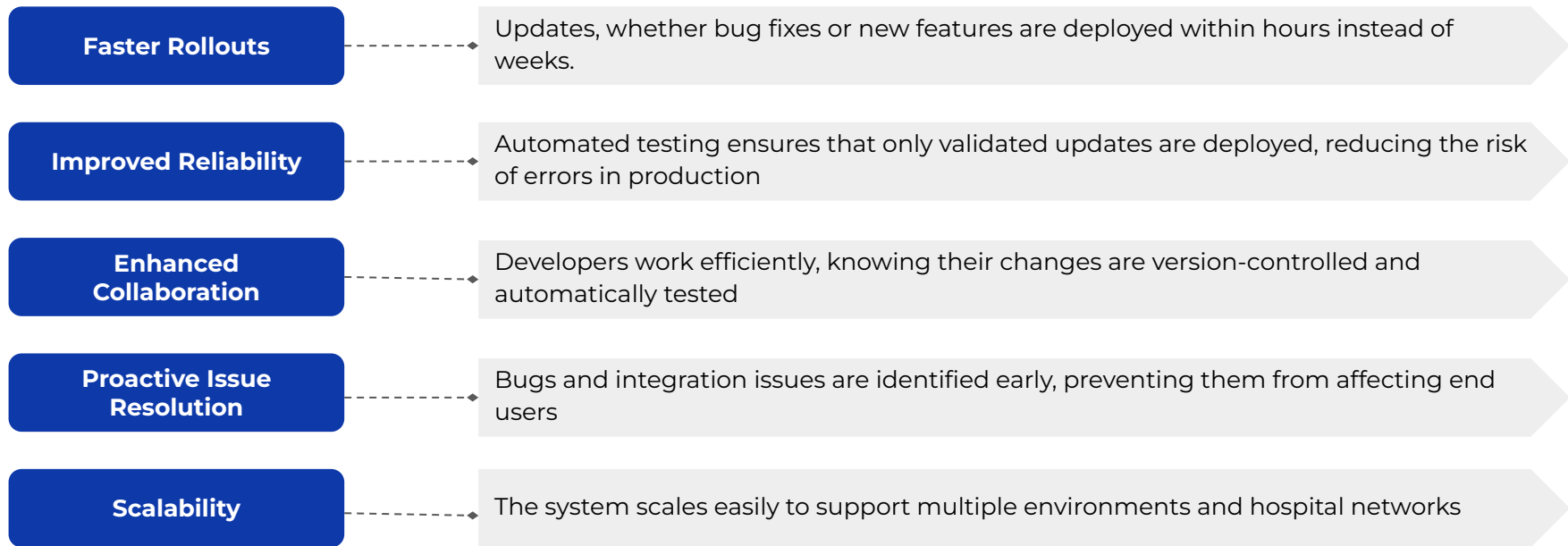
Enhanced Collaboration

Developers work efficiently, knowing their changes are version-controlled and automatically tested

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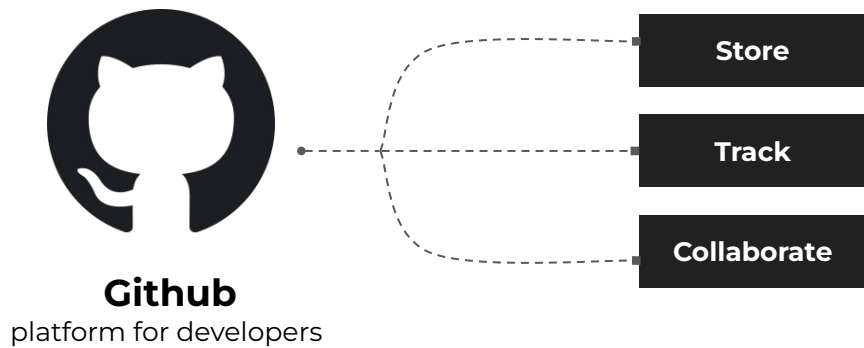
Introduction to Github



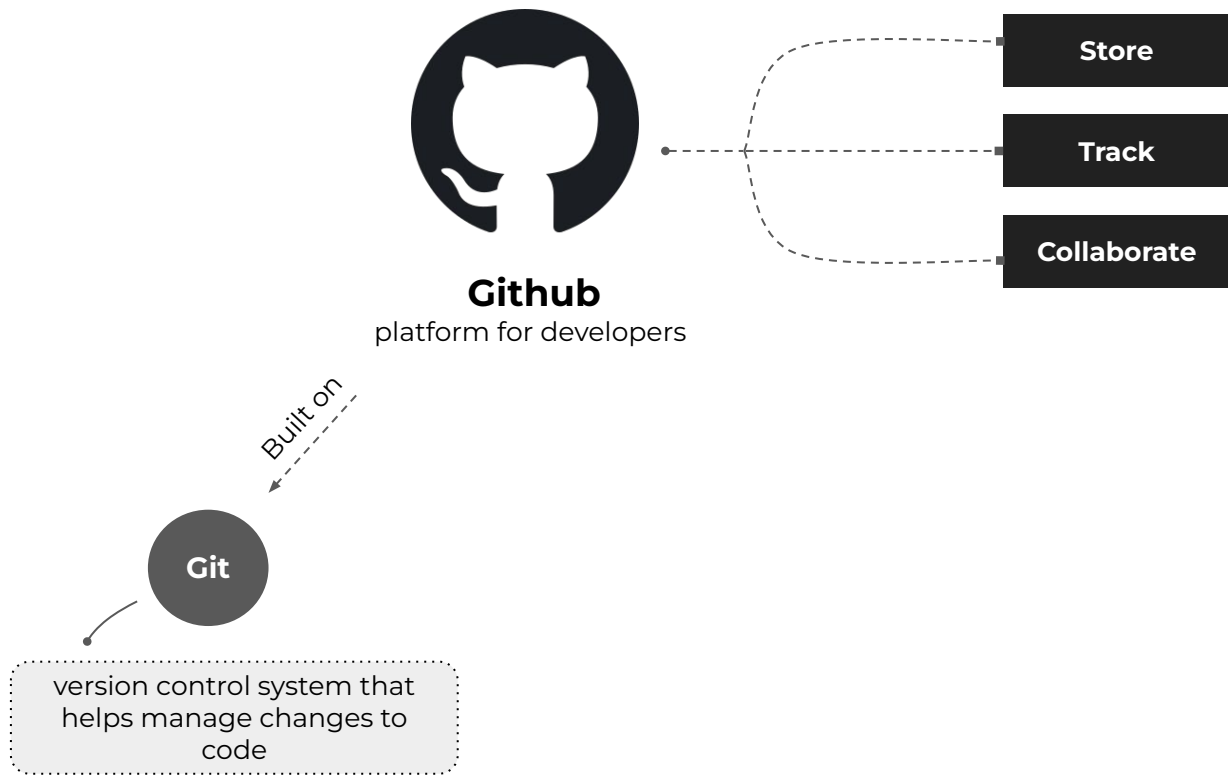
Github

platform for developers

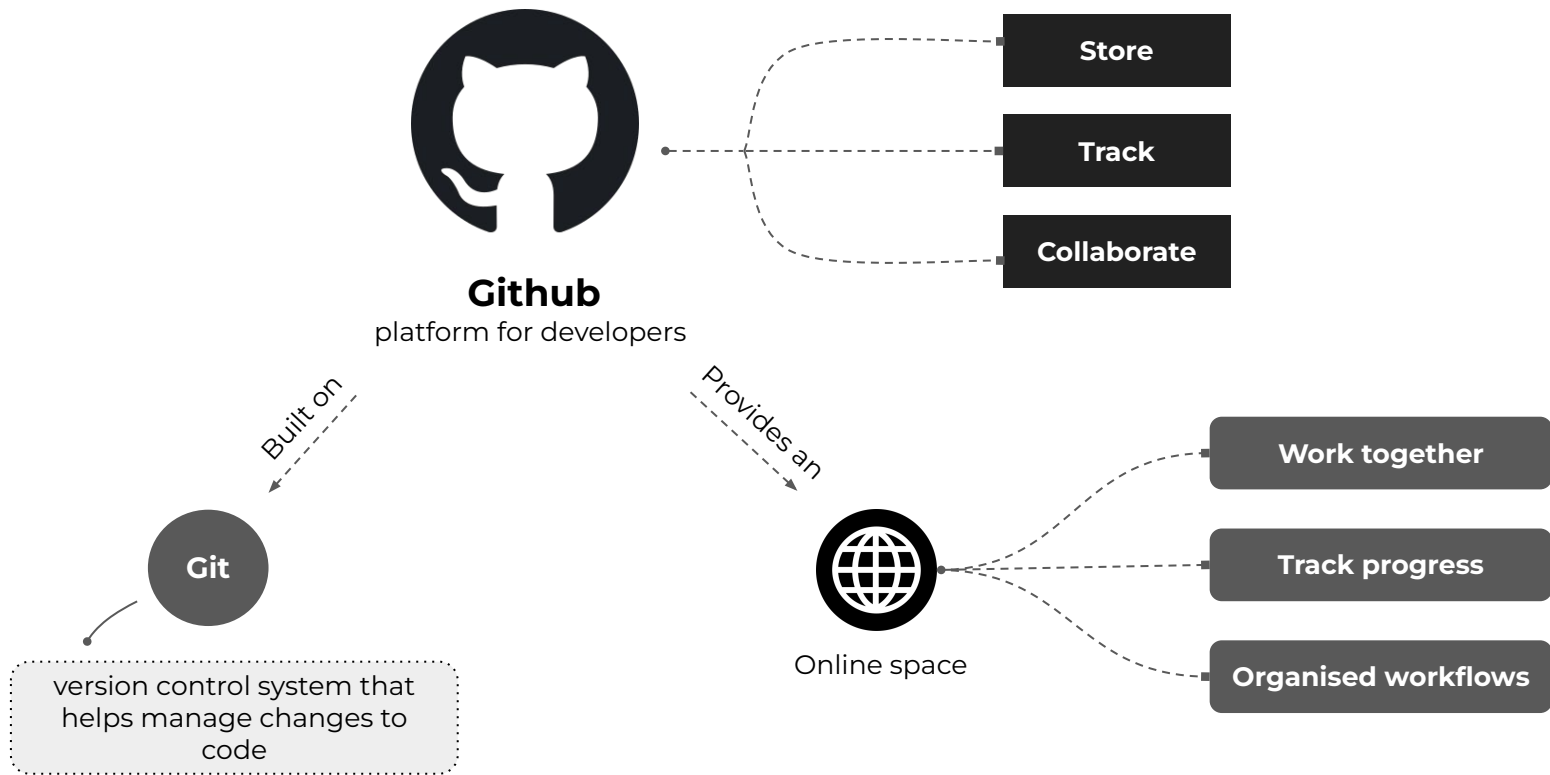
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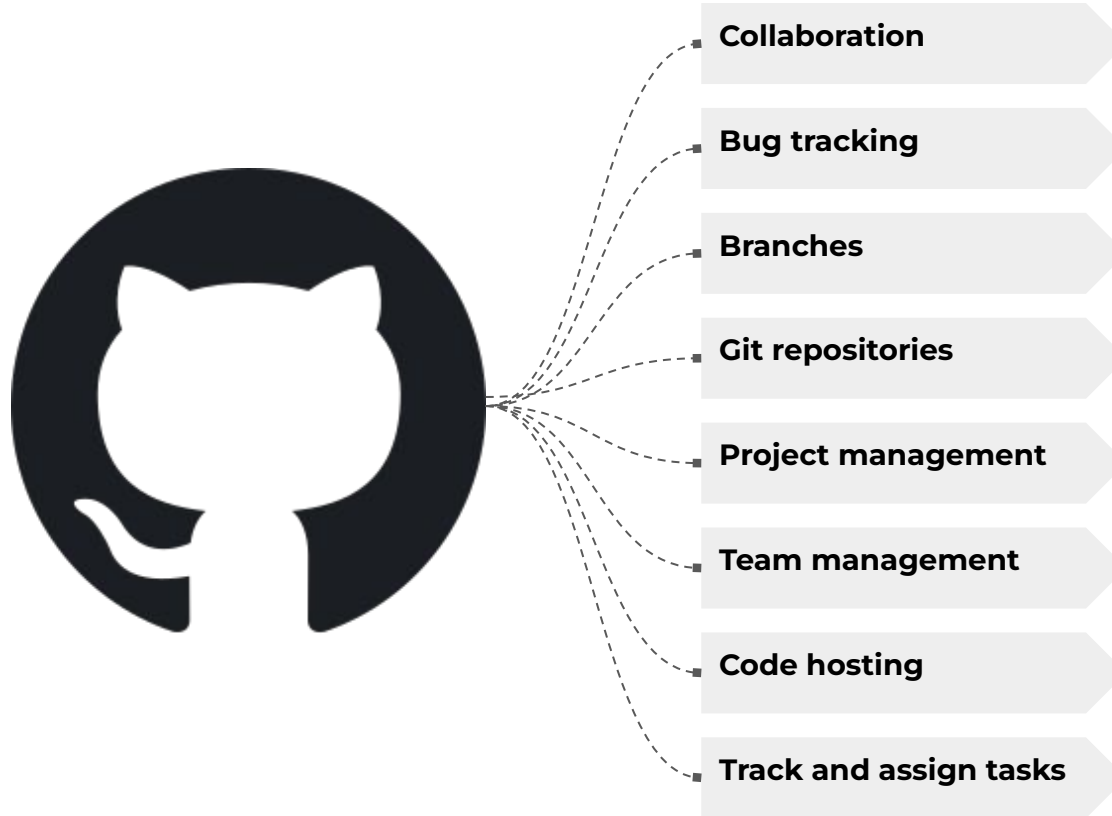
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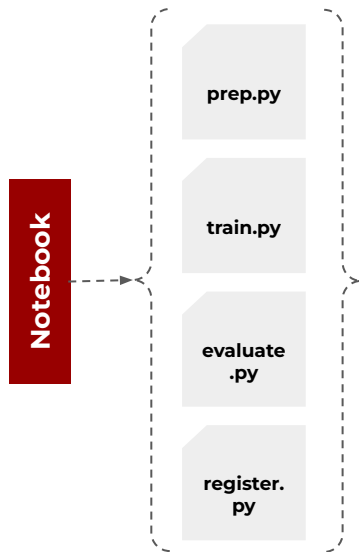
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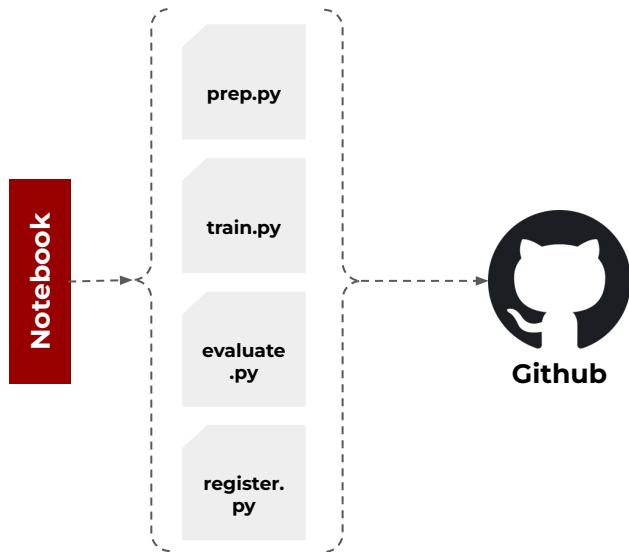
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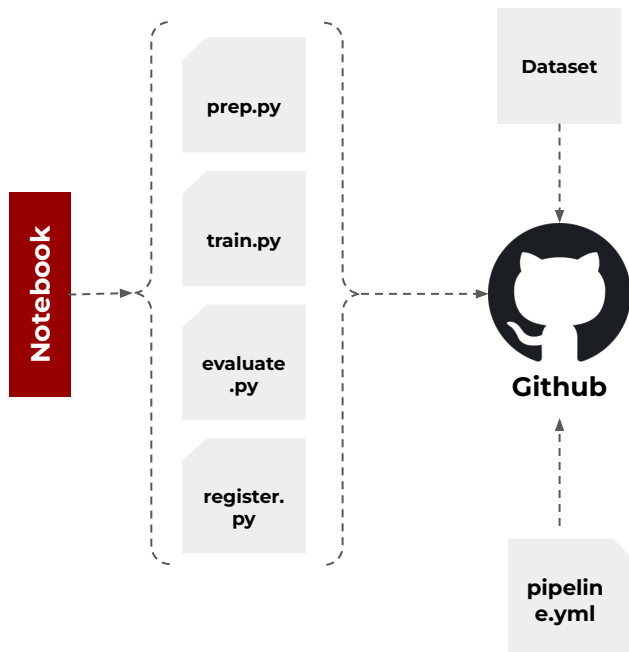
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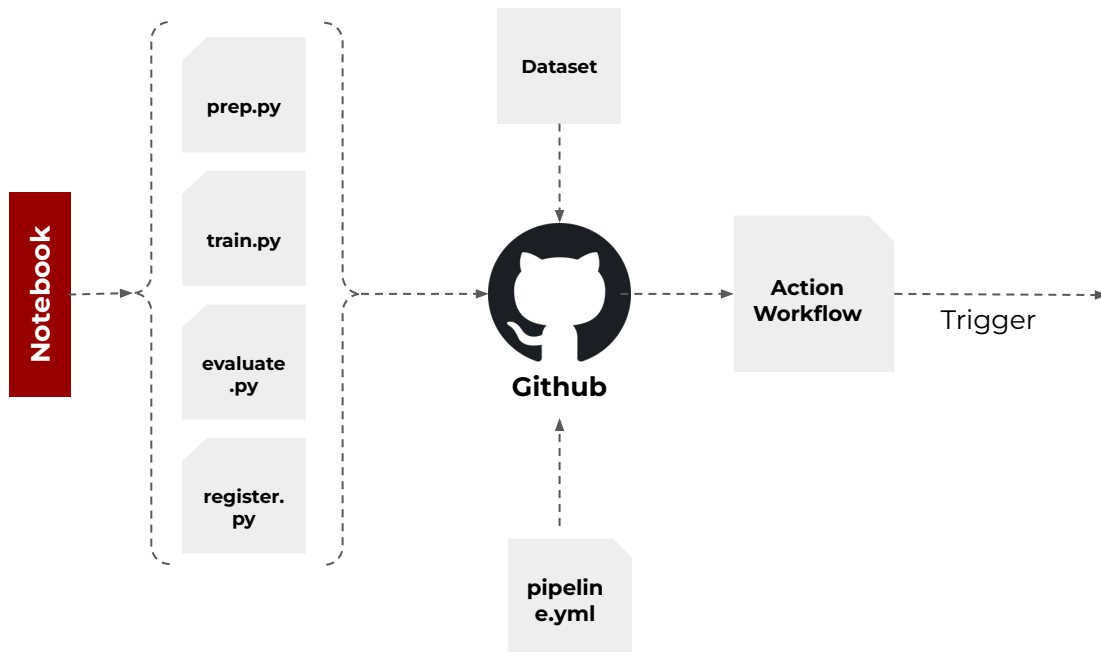
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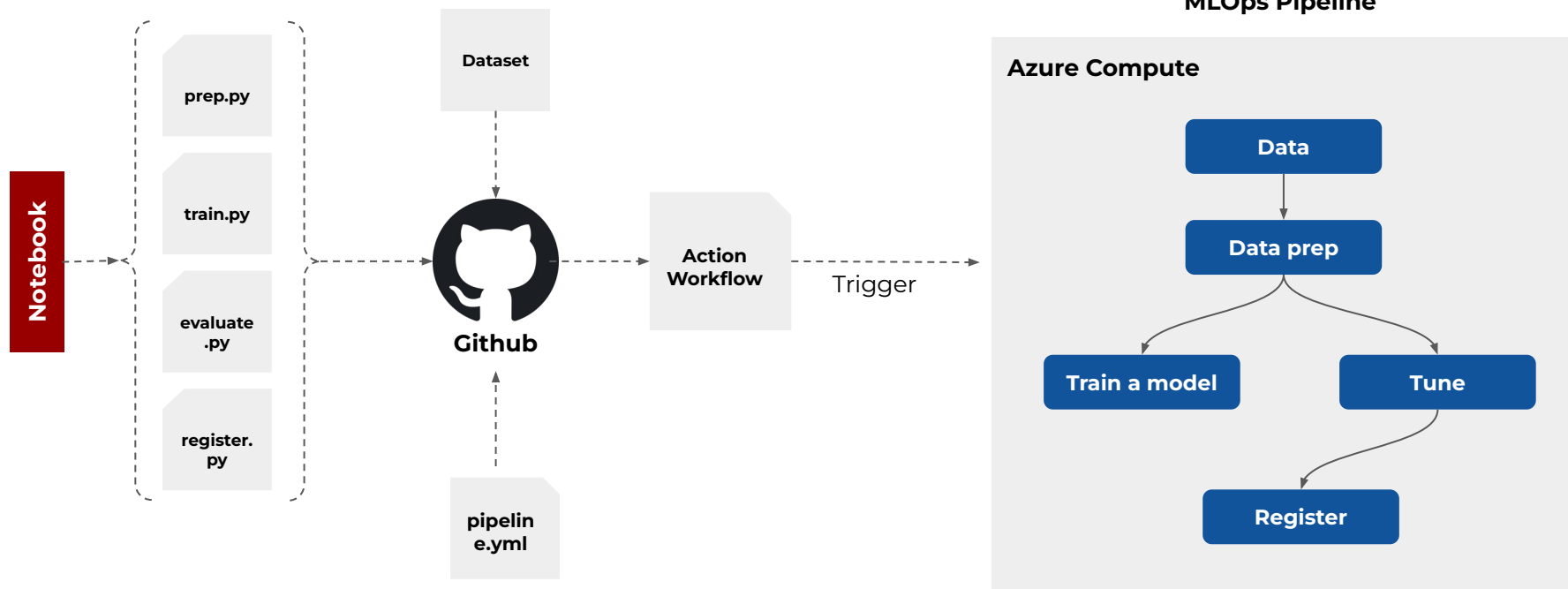
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Conclusion

- By adopting CICD, Nova ensures the system evolves rapidly to meet user needs while maintaining the stability and reliability critical in healthcare analytics.
- This scalable solution positions Nova as an innovator, trusted by healthcare providers for delivering timely, accurate insights.



Power Ahead!

